Week 3, Task 1 — Corpus Retention and Defect Audit

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Corpus audited: `problems_leetcode.jsonl`, run `06d15ab6-1127-431a-8c33-20f53d82a5ac`, collected 2026-07-17. 3,991 records.

> All counts below were produced by a script run over every record, not by inspection. No corrections have been applied and no re-collection has been started.

(a) Retention status

Raw responses were retained, embedded per record.

The collector did not write separate raw-response files. It stored the pre-parse form inside each record:

Field	Coverage	What it preserves
raw_metadata	3,991 / 3,991 (100%)	GraphQL response metadata as received
content_html	3,991 / 3,991 (100%)	Problem statement as served, pre-conversion

Consequence: **every parsing defect found below is correctable offline.** Re-deriving content_markdown, examples, constraints, or hints requires no network access, because the source form of each is still held. No re-collection needs to be scheduled for any parsing defect. The one exception is images, addressed in (c).

(b) Defect counts

Field completeness

Field	Empty	Of which premium	Residual
statement (content_markdown)	776 (19.4%)	776	0
constraints	900 (22.6%)	≤ 776	~124
examples	781 (19.6%)	≤ 776	~5
tags	124 (3.1%)	—	124
hints	1,610 (40.3%)	≤ 776	~834
difficulty	0	—	0

776 records are `paid_only` / `content_status=premium`; LeetCode does not serve their content. The statement row matching that figure exactly confirms the parser drops nothing on complete records.

Residuals are stated as approximate because they are arithmetic differences, not set intersections; exact residuals to be confirmed by set subtraction. Absent hints is expected on most problems and is likely not a defect.

Statement conversion — superscripts

> **2,145 records (53.7%) have corrupted exponents in content_markdown.** The HTML-to-markdown conversion strips <sup> tags and concatenates the exponent into the base. 11,340 <sup> occurrences corpus-wide.

Example (`two-sum`):

```
content_html     2 <= nums.length <= 10<sup>4</sup>
content_markdown 2 <= nums.length <= 104
```

10^4 becomes 104; 10^9 becomes 109; $O(n^2)$ becomes O(n2). Constraint magnitudes are misstated by orders of magnitude wherever they appear in the statement body.

The separate constraints array is unaffected — it holds the exponent correctly. The defect is confined to the markdown conversion path, and content_html is retained, so this is a pure offline re-derivation.

Invisible and non-ASCII characters

Character	Occurrences	Records
Zero-width space (U+200B)	2,203	182
Non-breaking space (U+00A0)	3,122	592
Rightwards arrow	943	53
Comparison operators	37	3
Smart quotes	21	11

Zero-width and non-breaking spaces carry a specific risk for the judge: they are not visually distinguishable from ordinary whitespace, and inside `examples` they would cause the `exact` comparison mode to reject correct submissions.

> **Checked: 0 records carry zero-width or non-breaking spaces inside the examples array.** All occurrences are confined to statement prose. The exact comparator is not exposed, and Task 5 can generate expected outputs from examples without sanitisation. Statement-prose occurrences remain cosmetic and correctable offline.

Rightwards arrows may be legitimate content in linked-list problems (1 -> 2 -> 3) rather than a defect. Not yet distinguished.

Other markup residue

Type	Occurrences	Records
Backslash escapes	47,274	2,464 (61.7%)
LaTeX fragments	232	22
Residual HTML tags	44	3
HTML entities	18	1

The backslash count is not yet confirmed as a defect. Literal escape sequences inside fenced code blocks may be intended content. Requires sampling before classification.

Example input/output count mismatch

0 mismatched, 0 unparseable. Every examples array has matching input and output counts. The exclusion criterion covering this case in the Sprint 1 strategy document did not need to fire on this run.

Images

730 records (18.3%) depend on at least one image; 1,468 references total. **1,466 are hot-linked to remote URLs; only 2 are local paths.** No image files are stored in the corpus.

A further 358 records (9.0%) are diagram-suggestive by tag or title but carry no image reference. These need manual spot-checking to distinguish problems that genuinely have no diagram from diagrams silently dropped during parsing.

Duplicates, identifier uniqueness, and provenance

Check	Result
Records carrying uid	3,991 / 3,991
Duplicate uid values	0
Duplicate slug values	0
Identical statement bodies	0
canonical_url present	3,991 / 3,991 (100%)
collected_at present	3,991 / 3,991 (100%)

(c) Findings sorted

Correctable offline

Every parsing defect falls here, because content_html and raw_metadata are retained.

1. Superscript corruption in content_markdown — 2,145 records
2. Zero-width and non-breaking space contamination — 592 records
3. Residual HTML tags in hints and elsewhere — 3 records
4. LaTeX fragments — 22 records
5. HTML entities — 1 record
6. Empty constraints / examples / tags on non-premium records — ~130 records
7. Backslash escapes — 2,464 records, pending classification

Requires re-collection

1. **Image assets — 1,466 hot-linked URLs across 730 records.** The corpus holds URLs, not files. If the source CDN paths change or rate-limit, these become unrecoverable. This is the only genuine re-collection exposure in the corpus and the item most at risk from delay.
2. Possible dropped diagrams — 358 records, pending spot-check. May resolve to zero.

Recommendation

The image download is the only work here that gets harder with time; everything else is recoverable from data already held. Suggest scheduling it as a standalone task rather than inside this week's remaining scope, since it competes directly with Tasks 3–5.

> The invisible-character risk to the judge has been checked and ruled out: no zero-width or non-breaking spaces appear inside the examples array. Task 5 is not blocked by this finding.